
Algorithmic Accountability in Small Data: Sample-Size-Induced Bias within Classification Metrics

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Abstract

Evaluating machine learning models is crucial not only for determining their technical accuracy but also for assessing their potential societal implications. While the potential for low-sample-size bias in algorithms is well known, we demonstrate the significance of sample-size bias induced by combinatorics in classification metrics. This revelation challenges the efficacy of these metrics in assessing bias with high resolution, especially when comparing groups of disparate sizes, which frequently arise in social applications. We provide analyses of the bias that appears in several commonly applied metrics and propose a model-agnostic assessment and correction technique. Additionally, we analyze counts of undefined cases in metric calculations, which can lead to misleading evaluations if improperly handled. This work illuminates the previously unrecognized challenge of combinatorics and probability in standard evaluation practices, hoping to advance the community’s approach for performing equitable and trustworthy classification methods.

1 INTRODUCTION

Classification metrics derived from confusion matrices are fundamental tools in evaluating machine learning models, particularly in binary classification tasks. Metrics such as accuracy, precision, recall, and Matthews Correlation Coefficient (MCC) provide critical insights into model performance. However, an

often-overlooked issue is the impact of sample size on these metrics, especially when comparing performance across different subgroups or datasets of varying sizes.

Small sample sizes introduce jaggedness (significant variability with discrete “jumps” in the score distribution) in classification metrics due to the discrete and combinatorial nature of confusion matrices. This variability can lead to misleading interpretations of a model’s performance and fairness. For instance, minor changes in the counts of true positives or false negatives can cause disproportionate shifts in metric values, complicating comparisons across groups and potentially obscuring biases or disparities.

Moreover, certain metrics become undefined under specific conditions, such as divisions by zero. These undefined cases, or “holes”, in the metric space further challenge the reliability of performance assessments, especially in subgroups with limited data.

Despite the widespread use of confusion-matrix metrics, there has been limited attention to the systematic biases and inconsistencies induced by sample size variations. Existing literature often assumes large sample sizes or overlooks the combinatorial complexities that small samples introduce, leaving a gap in understanding how to accurately assess model performance.

We address these challenges by providing a comprehensive analysis of sample-size-induced biases in confusion-matrix metrics. Our contributions include:

- **Demonstrating Metric Variability:** We present both theoretical and empirical evidence of the jaggedness and variability in classification metrics caused by small sample sizes. By illustrating how these effects distort performance evaluations, we underscore the need for careful interpretation of metrics in small-sample scenarios.
- **Quantifying Undefined Cases:** We systematically count the number of “holes”—situations where metrics are undefined—in several commonly used classification metrics. This quantifi-

cation reveals the extent to which certain metrics may be unreliable or misleading under specific sample configurations.

- **Metric Alignment Trial for Checking Homogeneity (MATCH) Test:** We introduce a statistical test that assesses the significance of a subgroup’s metric score by comparing it to the distribution of scores from a reference group. This approach allows us to determine whether observed differences are due to genuine performance disparities or merely artifacts of sample size variability.
- **Cross-Prior Smoothing (CPS):** We propose a smoothing technique that incorporates prior information from other groups to enhance metric reliability. By adjusting confusion matrix counts using cross-group priors, CPS uniformly reduces variability and improves the stability of metric estimates.
- **Open-Source Implementation:** We provide an open-source repository of our code and experiments to facilitate reproducibility and encourage further research in this area.¹

Our work illuminates a critical issue with classification metrics and offers practical solutions to improve their reliability. By addressing the biases introduced by sample size, we aim to enhance the fairness and accuracy of model evaluations, particularly in applications where subgroup comparisons are essential.

2 RELATED WORK

As machine learning models are increasingly used in high-stakes domains, ensuring fairness across subgroups is critical. However, the impact of sample size on classification metrics such as accuracy, precision, and recall remains underexplored. This study examines how these metrics evolve with changing sample sizes, revealing potential biases that may exacerbate disparities between subgroups. Understanding these distributional shifts can help develop more equitable AI systems, particularly in settings where sample sizes differ significantly. Metric definitions are provided in Section B.

Confusion matrices are fundamental tools for assessing classification performance, providing the basis for scalar metrics like accuracy and MCC. While these metrics are widely used, they can exhibit biases related to sample size—a phenomenon we refer to as *score distribution shift*. This bias affects common fairness metrics, including disparate impact (Feldman et al., 2015),

equalized odds (Hardt et al., 2016), and predictive parity (Das et al., 2021). By highlighting this issue, we aim to enhance the robustness of metric comparisons across datasets with varying sample sizes.

In social data analysis, accurate measurement of fairness is crucial, particularly in legal contexts where metrics like disparate impact assess potential discrimination. For instance, in employment or housing discrimination cases (Dennis J. Aigner and Wiles, 2024; Gastwirth and Miao, 2009) and recidivism prediction (Dressel and Farid, 2018; Moore et al., 2023), datasets often feature small or imbalanced sample sizes. Recognizing how distributional shifts in metrics affect evaluations is vital for ensuring fair outcomes. The recently introduced Objective Fairness Index (Briscoe and Gebremedhin, 2024) is also susceptible to such shifts.

Recent work by Feng et al. (2024) introduces a method to assess model calibration across subgroups, ensuring that models are reliable for different populations, particularly in high-dimensional settings with limited data.

Statistical methods designed for small sample sizes are relevant to our approach. Techniques like Laplacian smoothing, commonly used in natural language processing to handle unobserved events (Eisenstein, 2019), can be adapted to confusion matrices where certain classes may not be represented. Discussions around the use of non-informative or weakly informative priors in Bayesian estimation for small samples (He et al., 2021; Lemoine, 2019) inform our methodology, particularly in developing robust methods under data limitations.

Sokolova and Lapalme (2009) provides a comprehensive analysis of various performance measures used in classification tasks, focusing on their invariance properties with respect to changes in the confusion matrix. Their work introduces the concept of “measure invariance”, which refers to a metric’s ability to maintain consistent evaluations despite transformations in the underlying data distribution.

Additionally, Goutte and Gaussier (2005) introduces a probabilistic framework that interprets precision, recall, and the F_1 score by using a symmetric beta distribution and Monte Carlo sampling techniques. This approach moves beyond simple point estimates by estimating the likelihood that one algorithm would outperform another based on these scores. Part of our work extends this idea by generalizing precision and recall to a broader set of metrics, grouped as *Joint Ratio Metrics*.

The issue of metric unreliability in small sample sizes is often overlooked, with practitioners sometimes at-

¹<https://github.com/jarrenbr/Algorithmic-Accountability-in-Small-Data>

tributing it to the need for more data. Chicco and Jurman (2020) specifically addresses this issue along with imbalanced data by comparing MCC against the more widely used accuracy and F_1 -score. They argue that MCC provides a more reliable assessment of classifier performance in binary tasks, particularly when there is a class imbalance because MCC incorporates all four confusion matrix categories. In contrast, accuracy and F_1 score can give misleadingly high values in imbalanced datasets, failing to capture poor performance in one class.

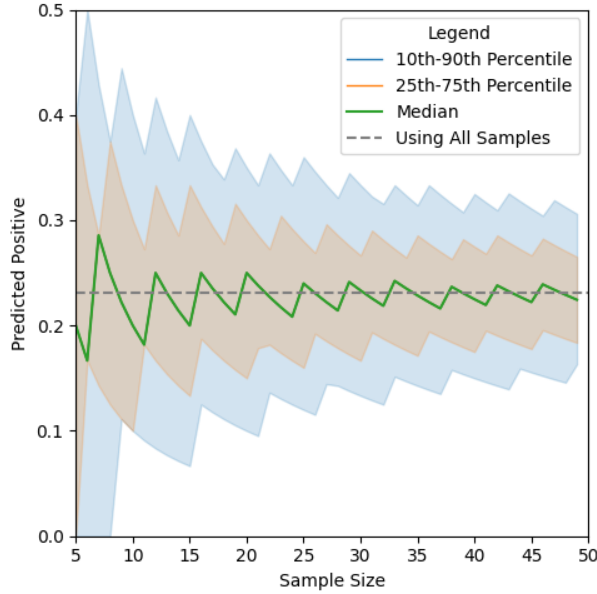


Figure 1: Variability of Positive Predictive Rate for Wealth Classification Among Multiracial Individuals: A Monte Carlo Simulation Study Based on Sample Size.

In another work, Rudner et al. (2024) adjusts model parameters with Group-Aware Priors to improve model robustness under subpopulation shifts. Aghbalou et al. (2024) examines imbalanced classification and derives sharp error bounds under class imbalance. Their focus is on improving classification performance when one class is underrepresented, offering new theoretical insights on error rates when class probabilities approach zero.

Our work builds on these foundations by examining the effect of small sample sizes on a broader range of classification metrics. We demonstrate that metrics like accuracy, precision, recall, and even MCC not only exhibit increased variability but also have discrete jumps and falls (jaggedness) as the sample size scales, potentially leading to misleading evaluations. For instance, as Chicco and Jurman present, while MCC is robust in imbalanced scenarios, it can still fluctuate

in extreme cases of imbalance or small samples, an issue our Cross-Prior Smoothing (CPS) technique aims to mitigate by using prior information from reference groups. Additionally, our introduction of the MATCH Test allows for a rigorous assessment of whether observed metric scores in small subgroups are meaningful or merely artifacts of sampling variability.

By addressing these gaps, we contribute to the growing literature on improving the robustness and fairness of classification metrics. Our approach offers both theoretical insights and practical tools to ensure more reliable evaluations across diverse subgroups, particularly when sample sizes are limited or imbalanced.

3 DISCRETE DISTRIBUTION SHIFTS

The variability and jaggedness in the predicted positive rates, as illustrated in Figure 1, is a direct consequence of both the probabilistic nature of small sample sizes and the combinatorial explosion of the confusion matrix space. This section explains the underlying phenomena contributing to these effects through a probabilistic lens.

Let us begin by considering the structure of the confusion matrix CM in Definition B.1.

Definition 3.1. *The binary confusion matrix, CM, is a 2×2 matrix of the non-negative counts of true positives (TP), false negatives (FN), false positives (FP), and true negatives (TN). Each of these counts is referred to as a cell c .*

$$CM \triangleq \begin{bmatrix} TP & FN \\ FP & TN \end{bmatrix} \quad (1)$$

The total number of samples n is partitioned among these four categories, such that $n = TP + FN + FP + TN$. For a given sample size n , we denote the set of all possible confusion matrices as $\mathcal{M}(n)$.

The narrowing of the distribution observed in Figure 1 can be attributed to the combinatorial expansion of $\mathcal{M}(n)$ with increasing n . For small sample sizes, the number of possible confusion matrices is limited, causing each configuration to represent a significant fraction of the total space, including those yielding extreme metric values. As n grows, the number of possible confusion matrices increases rapidly, and $\mathcal{M}(n)$ becomes densely populated. Specifically, the cardinality of $\mathcal{M}(n)$ is given by $|\mathcal{M}(n)| = \binom{n+3}{3}$ using the stars and bars method. Hence, the number of possible confusion matrices grows cubically with n .

These configurations are influenced by the underlying probability distribution of classification outcomes, denoted as $(p_{TP}, p_{FN}, p_{FP}, p_{TN})$. Using k_i as the integer

count of i , we use the multinomial distribution definition to define the probability of any CM configuration:

$$P(CM) = \frac{n!}{k_{TP}! k_{FP}! k_{FN}! k_{TN}!} p_{TP}^{k_{TP}} p_{FP}^{k_{FP}} p_{FN}^{k_{FN}} p_{TN}^{k_{TN}}. \quad (2)$$

Assumption 3.1. *Using the multinomial distribution assumes that 1) the data is sampled i.i.d., and 2) that the model’s classifications are independent.*

Due to the factorial terms in the multinomial coefficient from Equation 2, even a single-unit increase in n can significantly disrupt the probability mass distribution for small sample sizes, as depicted in Figure 1.

The score distribution shifts are not solely a probabilistic artifact. As n increases, the discrete space of possible confusion matrices becomes more finely granulated, and the number of configurations corresponding to certain metric scores increases at different rates. Figure 2 shows how the distribution of metric scores changes with varying n , as all possible confusion matrix configurations are considered for each group size.

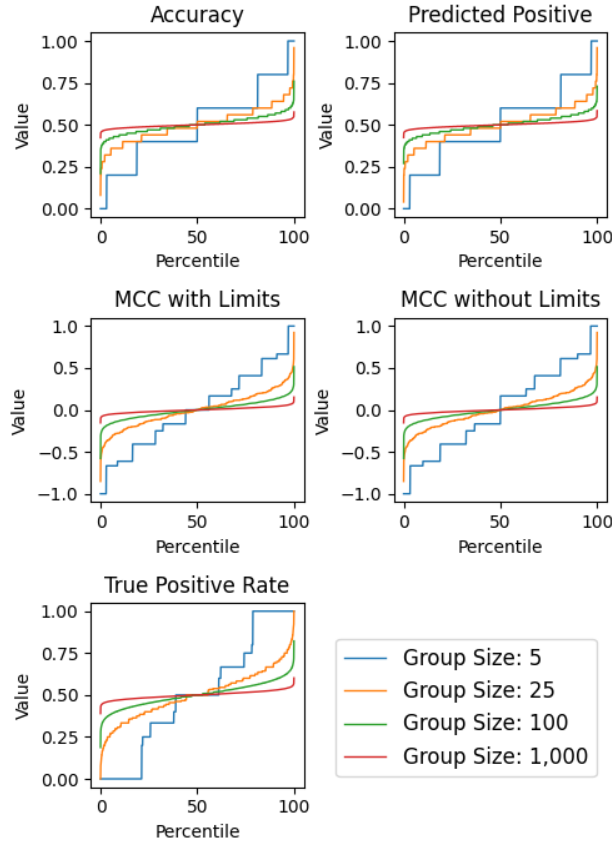


Figure 2: Distribution Shift in Classification: Analyzing Metric Sensitivity to Sample Size with the CDF.

As the sample size increases, the probability masses

of metric scores tend to converge around their expected values. In Figure 1, the expected value of the predicted positive rate converges to $1365/5956 \approx 0.23$. For any metric M , the expected value is $\mathbb{E}[M(p_{TP}, p_{FN}, p_{FP}, p_{TN})]$.

For example, Figure 2 uses all configurations, so all probabilities are 0.25. We see that accuracy and predicted positive converge to $(0.25+0.25)/1=0.5$; MCC (with or without limits) converges to $\frac{0.25 \cdot 0.25 - 0.25 \cdot 0.25}{\sqrt{(0.25+0.25) \cdot (0.25+0.25) \cdot (0.25+0.25) \cdot (0.25+0.25)}} = 0$; and TPR goes to $0.25/(0.25+0.25)=0.5$.

By understanding discrete distribution shifts, we highlight the importance of considering sample size effects when interpreting classification metrics, particularly where small subgroups are compared. Ignoring these effects can lead to misleading conclusions about performance and fairness across different populations.

4 UNDEFINED CASES

Classification metrics derived from confusion matrices can exhibit undefined behaviors, or “holes”, under certain conditions—particularly when denominators in their formulas become zero. In Table 1, we find that for all considered metrics, the asymptotic growth of undefined cases is less than the cubic growth of total possible configurations. We provide the proofs in Section C.

We generalize metrics of the form $M = \frac{c_i + c_j}{n}$ as *Binomial Metrics*, which include common measures like accuracy, prevalence, and predicted positive rate. Furthermore, we categorize metrics with form $M = \frac{c_i}{c_i + c_j}$, as *Joint Ratio Metrics* (JRM), which include common measures like true positive rate (TPR), false positive rate (FPR), precision (PPV), and others.

We also consider other metrics, including the new *Marginal Benefit*, which quantifies the net benefit or cost to a group (Briscoe and Gebremedhin, 2024). See Section B for a detailed definition of all 21 metrics considered.

5 MATCH TEST

The inherent variability and jaggedness in classification metrics due to small sample sizes complicate the assessment of model performance across groups of different sizes. Small variations in the sample size n can lead to disproportionately large changes in metric values, making direct comparisons potentially misleading. This issue is particularly acute in fairness evaluations, where comparing metrics across subgroups (e.g., different demographic groups) is essential.

Table 1: Undefined Cases Across Metrics Compared to Total Configuration Counts for all $n \geq 3$.

Metric	Asymptotic Growth	Undefined Case Count
Binomial Metrics (ACC, PREV, ...)	$\Theta(1)$	0
Marginal Benefit	$\Theta(1)$	0
Objective Fairness Index	$\Theta(1)$	0
Simplified F ₁ Score	$\Theta(1)$	1
Joint Ratio Metrics (TPR, FPR, ...)	$\Theta(n)$	$n + 1$
Matthews Correlation Coefficient	$\Theta(n)$	$4n$
Prevalence Threshold	$\Omega(n) \ \& \ \mathcal{O}(n \log \log n)$	$\geq 2n + 2 \text{ and } < e^\gamma n \log \log n + \frac{0.6483n}{\log \log n}$
Treatment Equality	$\Theta(n^2)$	$\binom{n_1+2}{2} + \binom{n_2+2}{2} - 1$
Original F ₁ Score	$\Theta(n^2)$	$\binom{n+2}{2}$
Unique Confusion-Matrices Count	$\Theta(n^3)$	Count of All Possible CMs: $\binom{n+3}{3}$

To address this challenge, we propose a *Metric Alignment Trial for Checking Homogeneity (MATCH) Test* that quantifies the likelihood of observing a given metric score under a reference distribution. Specifically, we assess whether an observed metric score, $M(\text{CM}_i)$, computed from a confusion matrix CM_i for subgroup i , is consistent with what would be expected if this subgroup followed the same performance distribution as a reference group. By comparing the observed score from subgroup i against the cumulative distribution function (CDF) derived from the reference group, we determine the percentile rank of the observed score within this distribution. This method allows us to evaluate the statistical significance of observed metrics, facilitating more informed interpretations when sample size variations could skew direct comparisons.

We focus on several classification metrics, including accuracy, prevalence, and predicted positive rate (collectively referred to as *Binomial Metrics*), as well as metrics like the *marginal benefit* $\mathcal{B}$, and JRMs. For each metric, we derive methods to compute the cumulative probability of an observed score S_{obs} under the assumption that it is drawn from the reference distribution.

To prepare, we now define several terms. Let $k_i \geq 0$ denote the integer count of c_i , p_i is the probability of c_i , $q_i = 1 - p_i$ is the probability of not c_i , S_{obs} is the observed score of a metric M , and $P(S \leq S_{\text{obs}}) \in [0, 1]$ is the probability that $S \leq S_{\text{obs}}$. The subscript $i + j$ means i or j .

5.1 Binomial Metrics

We find distributions for accuracy, prevalence, and predicted positive which have the form $(c_i + c_j)/n$.

Theorem 5.1. *For Binomial Metrics, the cumulative probability is given by the binomial cumulative distribution function (CDF):*

$$P(S \leq S_{\text{obs}}) = \sum_{f=0}^{k_{i+j}} \binom{n}{f} p^f q^{n-f} \quad (3)$$

Proof. The total count k_{i+j} corresponds to the number of successes in n trials, each with success probability p . Thus, k_{i+j} follows a binomial distribution with parameters n and p . $\square$

When $np \geq 5$ and $nq \geq 5$, we can approximate $P(S \leq S_{\text{obs}})$ using the normal distribution due to the Central Limit Theorem (Ye et al., 2024; Freund et al., 2014). This is done in constant time. By definition, the mean $\mu = np_{i+j}$ and standard deviation $\sigma = \sqrt{np_{i+j}q_{i+j}}$.

$$\text{Standardize: } z = \frac{k_{i+j} + 0.5 - \mu}{\sigma} \quad (4)$$

The $+0.5$ is the continuity correction as we approximate this discrete distribution with a continuous one. Then, we use the standard normal CDF with z : $\Phi(z)$. Furthermore, by the Berry-Esseen Theorem, the error is $\mathcal{O}(n^{-1/2})$ (Berry, 1941; Esseen, 1942).

For smaller n , exact computation using the binomial CDF is feasible in $\mathcal{O}(n^2)$ time. Otherwise, if the approximation criteria do not hold, one can use efficient approximations such as the Lanczos approximation for $\mathcal{O}(n \log n)$ time (Lanczos, 1964).

Example. Consider $n=100$, $p=0.75$, and observed score $S_{\text{obs}}=0.80$. Then $k_{i+j} = nS_{\text{obs}} = 80$, $\mu = 75$, $\sigma \approx 4.33$, and $z \approx 1.27$. Thus, $P(S \leq 0.80) \approx \Phi(1.27) \approx 0.90$, so the observed score is higher than

approximately 90% of the expected outcomes under the reference distribution.

5.2 Marginal Benefit

The marginal benefit is defined as $\mathcal{B} = (\text{FP} - \text{FN})/n$, and is designed to measure the net benefit or cost to a group (Briscoe and Gebremedhin, 2024). Unlike Binomial Metrics, $\mathcal{B}$ involves the difference of counts, making its distribution symmetric around zero when $p_{FP} = p_{FN}$. We provide the CDF for $\mathcal{B}$ in Theorem D.2, and a theorem for approximating it using the normal distribution in Theorem 5.2.

Theorem 5.2. *The cumulative probability for $\mathcal{B}$ can be approximated using the normal distribution:*

$$F(S \leq S_{\text{obs}}) \approx \Phi \left(\frac{k_{FP} - k_{FN} - (p_+ - p_-)}{\sqrt{(p_+ + p_-) - (p_+ - p_-)^2}} \right) \quad (5)$$

Proof. $\mathcal{B}$ can be expressed as the average of n i.i.d. random variables X_i : $\frac{1}{n} \sum_{i=1}^n X_i$. For

$$X_i = \begin{cases} +1 & \text{with probability } p_+ = p_{FP} \\ -1 & \text{with probability } p_- = p_{FN} \\ 0 & \text{with probability } p_0 = p_{TN+TP}. \end{cases} \quad (6)$$

Now we calculate the moments of $\mathcal{B}$.

$$\mu = \mathbb{E}[X_i] = p_+ - p_- + 0 \cdot p_0 = p_+ - p_- \quad (7)$$

For σ , we use the expanded definition: $\sigma^2 = \mathbb{E}[X_i^2] - \mu^2$.

$$\mathbb{E}[X_i^2] = (+1)^2 p_+ + (-1)^2 p_- + (0)^2 p_0 \quad (8)$$

$$\text{Thus, } \sigma^2 = (p_+ + p_-) - (p_+ - p_-)^2. \quad (9)$$

Then we calculate the z-score and use the normal CDF Φ . Note that the continuity correction δ cancels out: $z = ((k_{FP} + \delta) - (k_{FN} + \delta) - \mu) / \sigma$. $\square$

The Berry-Esseen Theorem also guarantees that this error is $\mathcal{O}(n^{-1/2})$ (Berry, 1941; Esseen, 1942).

5.3 Joint Ratio Metrics

Joint Ratio Metrics (JRM) are expressed as $c_i / (c_i + c_j)$. Computing the cumulative probability for JRM presents additional challenges due to the ratio of random variables involved. Theorem 5.3 extends the work of Goutte and Gaussier by introducing a generalized framework for these metrics, along with an analysis of computational complexity. For simplicity in the following discussion, we define $k_{i+j} = k$ and $p_{i+j} = p$.

Theorem 5.3. *The cumulative probability for a JRM score S_{obs} is given by:*

$$\begin{aligned} P(S \leq S_{\text{obs}}) &= \sum_{k=1}^n P(c_{i+j} = k) P\left(\frac{k_i}{k} \leq S_{\text{obs}} \mid c_{i+j} = k\right) \\ &= \sum_{k=1}^n \left(\binom{n}{k} p^k (1-p)^{n-k} \cdot \sum_{k_i=0}^{k_i^{\max}} \binom{k}{k_i} \theta^{k_i} (1-\theta)^{k-k_i} \right) \end{aligned} \quad (10)$$

Proof Sketch. First, c_{i+j} follows a binomial distribution with parameters n and p . Given $c_{i+j} = k$, c_i follows a binomial distribution with parameters k and $\theta = \frac{p_i}{p}$. Since $c_i \subseteq c_{i+j}$, $P(c_{i+j} | c_i) = 1$. Then, by Bayes' Rule, $\theta = P(c_i | c_{i+j}) = 1 \cdot p_i / p$. See Section E for a full proof.

Computing this sum directly has computational complexity $\mathcal{O}(n^2)$. However, we can improve efficiency by approximating the distribution of the JRM using the beta distribution. When applying a Bayesian approach with a uniform prior (equivalent to adding pseudo-counts $\lambda = 1$), the posterior distribution of the JRM is a beta distribution with parameters $k_i + \lambda$ and $k_j + \lambda$.

Using the beta distribution, the cumulative probability is approximated as:

$$P(S \leq S_{\text{obs}}) \approx I_{S_{\text{obs}}}(k_i + \lambda, k - k_i + \lambda) \quad (11)$$

where I is the regularized incomplete beta function. This improves the time complexity to $\mathcal{O}(n)$ since I integrates once over an independent variable. Beta tables allow constant time if they are available. Otherwise, efficient continued fraction and recursive approximations, or quadrature approximations of integrals may allow further improvements as the integrand is well-behaved.

6 CROSS-PRIOR SMOOTHING

We introduce *Cross-Prior Smoothing* (CPS), a novel correction method designed to enhance the reliability of CM metrics, particularly when comparing groups of varying sizes or distributions. CPS addresses the inherent variability and instability in classification metrics that arise due to sample size disparities while preserving the consistency of bias assessments across different subgroups.

CPS leverages information from a reference group's confusion matrix, CM_{ref} , to inform and smooth the metrics derived from a target group's confusion matrix, CM_i . This approach is especially pertinent in fairness analysis, where both CM_i and CM_{ref} are generated by the same underlying algorithm but may represent different demographic groups.

Let CM_{total} denote the confusion matrix derived from the entire dataset, and $CM_i \subset CM_{\text{total}}$ be the confusion matrix for subgroup i . In our experiments, we define the reference confusion matrix, CM_{ref} , as the portion of CM_{total} excluding the data from group i : $CM_{\text{ref}} = CM_{\text{total}} \setminus CM_i$.

Assumption 6.1. *The reference confusion matrix CM_{ref} provides a sufficiently informative prior for correcting the metrics derived from CM_i .*

As we demonstrate by our experiments, this assumption is natural in practice, particularly given that once the parameters are fixed for a given model, the multinomial parameters are simple to estimate.

We define CPS in Algorithm 1. We normalize CM_{ref} to prevent it from dominating the target group that has small sample sizes. Furthermore, the choice of λ is crucial for balancing the contribution of the reference group’s data. We set $\lambda = 5$ for all experiments. Our algorithm is inspired by the Dirichlet distribu-

Algorithm 1: Cross-Prior Smoothing (CPS)

Input: Confusion matrix for group i , CM_i ;
Normalized reference confusion matrix,
 $\hat{CM}_{\text{ref}}$; Smoothing constant λ
Output: Smoothed confusion matrix CM_{smooth}
Create the concentrations (α) using the prior;
foreach $c \in CM_i$, $c' \in \hat{CM}_{\text{ref}}$ **do**
 $\alpha_c = c + \lambda \cdot c'$;
foreach $c_{\text{smooth}} \in CM_{\text{smooth}}$ **do**
 Normalize the posterior distribution;
 $c_{\text{smooth}} = \alpha_c / (\sum_{\alpha} \alpha)$;
 Scale back for size-dependent metrics;
 $c_{\text{smooth}} \leftarrow c_{\text{smooth}} \cdot |CM_i|$;
return Smoothed confusion matrix CM_{smooth} ;

tion, which is a conjugate prior to the multinomial distribution. However, this approach extends beyond a basic Bayesian estimate with a non-informative prior. By incorporating a reference group’s data, the CPS technique provides a more robust estimate, particularly when the sample size n is small. This smoothing method significantly improves the stability and reliability of metric estimates by reducing the variability and jaggedness that arise in small-sample scenarios.

6.1 Experiments and Results

We evaluate CPS across a range of commonly used classification metrics, including the eight Joint Ratio Metrics (JRM), accuracy, prevalence, predicted positive rate, marginal benefit, MCC, F_1 score, and prevalence threshold. Our experiments demonstrate that

CPS consistently improves the reliability of these metrics, as evidenced by reductions in both mean-absolute error (MAE) and mean-squared error (MSE). For empirical validation, we conduct experiments using two fairness datasets: COMPAS and the Folktale income dataset. As datasets commonly used in the fairness literature, these are powerful examples since they have realistically divergent CMs.

COMPAS Dataset: The COMPAS dataset comprises risk assessments used in the U.S. judicial system, which have been scrutinized for potential bias (Angwin et al., 2016; Larson et al., 2016; ProPublica, 2016). We utilize the confusion matrices reported in ProPublica to evaluate the effectiveness of CPS in a real-world fairness context.

Folktale Income Dataset: We also employ the Folktale income dataset, training a random forest classifier to predict whether individuals earn more than \$50k per year based on features such as marital status, race, and education. For bias assessment, we partitioned the data into eight racial groups and computed the corresponding confusion matrices.

To support the broad applicability of the data-driven prior, CPS, Table 2 highlights a problematic CM from the COMPAS dataset.

For each metric, we conduct ten subgroup to reference-group experiments, where we Monte-Carlo sample one million times for every sample size, 5 to 150. This results in **1.45 billion samples per metric**. Remarkably, CPS improves the MAE and MSE over the baseline in every experiment. Figure 3 illustrates the reduction in MSE for MCC, FPR, TPR, and PT across all ten subgroups, with **95% confidence intervals that are nearly indistinguishable**.

Our improvement in MCC is particularly encouraging, as this metric is well-recognized for its strong informational value (Chicco and Jurman, 2023). When ranking metrics by the space between the performance curves, PT shows the highest improvement among the fifteen metrics analyzed, while FPR shows the least. For TPR, we notice a unique behavior, where its error rate rises steadily from $n=1$ until approximately $n=8$, before stabilizing. This is because of the small additive smoothing of $1e^{-10}$ applied to the baseline to ensure fair comparisons. Without the smoothing, TPR follows a monotonic, curved descent, starting at 0.035.

An additional advantage of CPS is its ability to mitigate issues arising from undefined metric values. By incorporating information from CM_{ref} , CPS drastically reduces the likelihood of encountering zero counts in corresponding cells of CM_i , which can lead to un-

Table 2: Comparison of the COMPAS dataset’s Black CM with Possible Prior Assumptions

Category	Normalized TP	Normalized FN	Normalized FP	Normalized TN
Black	0.09	0.33	0.05	0.53
All Other Races	0.06	0.25	0.05	0.64
Possible Prior Assumptions				
No Prior Assumption	0.00	0.00	0.00	0.00
CPS Estimate	0.06	0.25	0.05	0.64
Uniform (“Add 1”) Prior	0.25	0.25	0.25	0.25
Absolute Differences from Black CM				
No Prior Assumption	0.09	0.33	0.05	0.53
CPS Estimate	0.03	0.08	0.00	0.11
Uniform (“Add 1”) Prior	0.16	0.08	0.20	0.28

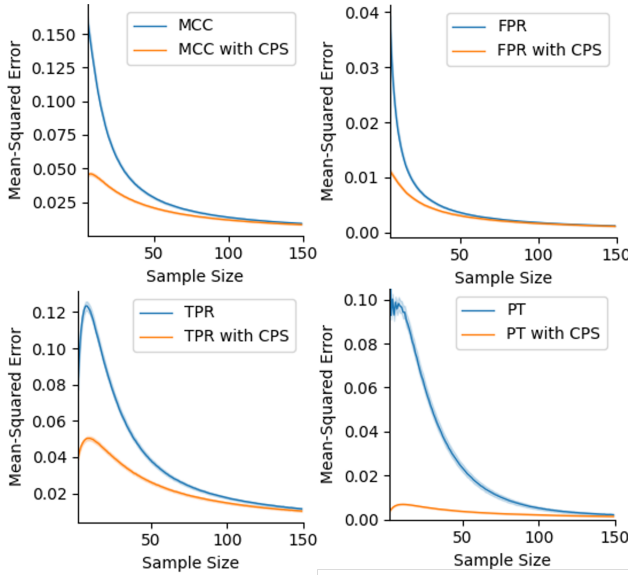


Figure 3: Sample-Size-Induced Bias Among all Groups: Impact of Cross-Prior Smoothing.

consistently closer to the observed, whole-group score.

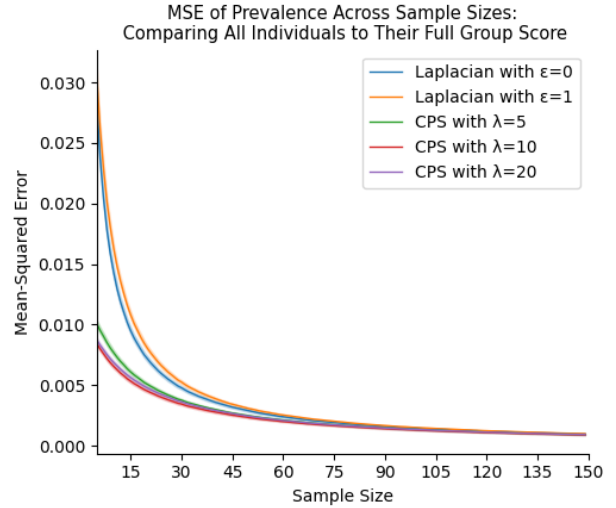


Figure 4: Comparing Cross-Prior Smoothing with the original metric, and the add-one technique.

defined scores.

In Figure 4, we focus on the prevalence metric to compare the original metric (Laplacian with $\varepsilon = 0$), adding one to each cell technique (Laplacian with $\varepsilon = 1$), and CPS with $\lambda \in \{5, 10, 20\}$. We see that CPS improves over all other techniques, with increasing the prior’s weight (λ) giving diminishing returns. $\lambda \geq 40$ tends to not perform as well as $\lambda = 10$.

In Section F.1, we provide several more experiments. There, we show that the adding one technique is inconsistent, sometimes performing worse than the original metric, and sometimes performing better than CPS. For all fifteen metrics tested with our down-sampling technique, CPS dominates the original metric, being

6.2 Bias-Variance Trade-Off

We investigate the bias-variance trade-off of CPS by analyzing the underlying multinomial distribution of CMs. The proof is provided in Section F.3.

Theorem 6.1. *Given the Cross-Prior Smoothing (CPS) estimate $\hat{p}_c^{CPS}$, the bias and variance have the following asymptotic behavior with respect to n and λ :*

$$\text{Bias}(\hat{p}_c^{CPS}) \in \mathcal{O}\left(\frac{\lambda}{\lambda + n}\right), \quad (12)$$

$$\text{Var}(\hat{p}_c^{CPS}) \in \mathcal{O}\left(\frac{n}{(n + \lambda)^2}\right). \quad (13)$$

Corollary 6.1.1. *Fixing n , the bias and variance of $\hat{p}_c^{CPS}$ with respect to λ satisfy:*

$$\text{Bias}(\hat{p}_c^{CPS}) \in \mathcal{O}(1), \quad \text{Var}(\hat{p}_c^{CPS}) \in \mathcal{O}\left(\frac{1}{\lambda^2}\right).$$

For a given n , CPS effectively addresses the *sample-size-induced bias* arising from the discrete, multinomial nature of confusion-matrix metrics (see Figure 1) by adjusting the smoothing parameter λ . Specifically, and with a constant n , as λ increases, the variance of each cell decreases on the order of $\mathcal{O}(1/\lambda^2)$, while the cell estimates are biased toward p'_c on the order of $\mathcal{O}(1)$. This behavior parallels well-known shrinkage approaches such as the James–Stein estimator, wherein one trades off smaller variance for additional bias. It is important to note, however, that this small-sample “bias”—an artifact of limited data and discrete counting—does *not* correspond to the classical bias term in the bias-variance trade-off for confusion-matrix cells. Instead, it reflects how some values of n can systematically shift metric scores away from their population-level score.

6.3 Discussion

CPS offers a practical and effective solution for correcting classification metrics in the presence of sample-size-induced bias. By using the reference group’s confusion matrix as a prior, we can achieve more stable and reliable metric estimates without compromising the integrity of bias assessments. However, it is important to note that the success of CPS hinges on the validity of Assumption 6.1. By experimenting with popular bias datasets, we provide supporting evidence of CPS’s broad applicability. Nevertheless, we recommend following this practical guide for CPS:

1. **Reasonable Prior Assumption:** CPS is most effective when the reference confusion matrix (the prior) is reasonably expected to be more informative than a uniform prior (e.g., adding one to each cell). Domain knowledge can often justify this; for instance, in the COMPAS dataset, we know that most criminals will not reoffend.
2. **Reference Size:** The reference group should have $n \geq 100$. This is due to the reference having sample-size-induced bias itself. Our experiments show there is significant bias for $n < 50$, and the bias becomes negligible once $n \approx 100$.
3. **Choosing λ :** Values in $[5, 10, 20]$ worked well empirically. Users can tune λ by iterative refinement.

Alternatively, methods such as synthetic stress tests (down-sampling and cross-validation), marginal distribution comparisons, and divergence measures (e.g.,

KL) can support CPS’ applicability to the given problem.

To evaluate CPS under divergent conditions, we analyze one of the most problematic CMs in the COMPAS dataset, whose systematic bias was publicized by ProPublica and has become a staple in the bias/fairness problems of ML.

7 CONCLUSIONS

In this paper, we present a comprehensive analysis of sample-size-induced bias in confusion-matrix metrics, highlighting significant implications for evaluating classification performance and fairness across groups of varying sizes. Our theoretical exploration revealed that small sample sizes lead to increased variability and jaggedness in metric scores due to the discrete and combinatorial nature of confusion matrices.

To address these challenges, we propose two novel approaches. First, we introduce the *MATCH Test* that assesses the statistical significance of an observed metric score relative to a reference distribution. This test enables more informed comparisons across groups by accounting for the variability introduced by different sample sizes. Second, we develop *Cross-Prior Smoothing* (CPS), a method that leverages prior information from a reference group’s confusion matrix to correct and stabilize metrics derived from smaller subgroups. CPS consistently improved the reliability of all 15 metrics tested, as evidenced by reductions in mean-absolute error and mean-squared error in our experiments on the COMPAS and Folktale income datasets.

Our findings underscore the importance of accounting for sample-size-induced bias when interpreting classification metrics, especially in fairness assessments where subgroup comparisons are critical. By revealing that sample size can be manipulated to present misleadingly favorable results, we highlight a vulnerability in current evaluation practices.

Future work could explore optimizing sample sizes to minimize metric variability and potential exploitation. Investigating the most advantageous configurations that could be misused and refining the MATCH Test by incorporating uncertainty in reference probabilities are also promising directions. Another avenue may explore behavior for different metrics; we have noticed that binomial metrics exhibit the most jaggedness. Additionally, extending the theoretical foundations of CPS and examining its applicability to a broader range of metrics and settings (such as multiclass classification) would further enhance its utility.

By providing both theoretical insights and practical

solutions, this work contributes to the development of more robust and equitable evaluation methods in machine learning, promoting fairer and more accurate assessments across diverse populations.

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Checklist

- For all models and algorithms presented, check if you include:
 - A clear description of the mathematical setting, assumptions, algorithm, and/or model. [Yes]
 - An analysis of the properties and complexity (time, space, sample size) of any algorithm. [Yes]
 - (Optional) Source code, with specification of all dependencies, including external libraries. [Yes]
- For any theoretical claim, check if you include:
 - Statements of the full set of assumptions of all theoretical results. [Yes]
 - Complete proofs of all theoretical results. [Yes—Some are relegated to the supplementary material]
 - Clear explanations of any assumptions. [Yes]
- For all figures and tables that present empirical results, check if you include:
 - The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). [Yes]
 - All the training details (e.g., data splits, hyperparameters, how they were chosen). [Yes]
 - A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). [Yes]

- (d) A description of the computing infrastructure used. (e.g., type of GPUs, internal cluster, or cloud provider). [Not Applicable]
- 4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:
 - (a) Citations of the creator If your work uses existing assets. [Not Applicable]
 - (b) The license information of the assets, if applicable. [Not Applicable]
 - (c) New assets either in the supplemental material or as a URL, if applicable. [Not Applicable]
 - (d) Information about consent from data providers/curators. [Not Applicable]
 - (e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. [Not Applicable]
- 5. If you used crowdsourcing or conducted research with human subjects, check if you include:
 - (a) The full text of instructions given to participants and screenshots. [Not Applicable]
 - (b) Descriptions of potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable. [Not Applicable]
 - (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. [Not Applicable]

Supplementary Material

A Counting Confusion Matrices

Here, we count the number of configurations (unique confusion matrices) possible given n in Theorem A.1 and the count of how many times cell c equals some value x given n in Theorem A.2.

Theorem A.1. *The cardinality of $\mathcal{M}(n)$ (the space of all possible confusion matrices given size n) is $\mathcal{N}(n) = \binom{n+3}{3} = (n+1)(n+2)(n+3)/6$.*

Proof. We prove this using the stars and bars method, which is defined as:

$$\binom{n^* + k - 1}{k - 1}, \quad (14)$$

where n^* represents the number of samples, and k denotes the number of buckets.

For our case, we have $n^* = n$ samples and $k = 4$ buckets, which corresponds to $k-1 = 3$ bars. Applying the formula, we obtain:

$$|\mathcal{M}(n)| = \mathcal{N}(n) = \binom{n+4-1}{4-1} = \frac{(n+1)(n+2)(n+3)}{6}. \quad (15)$$

□

Corollary A.1.1. *From Equation 15, the complexity of $\mathcal{N}(n)$ is $\mathcal{O}(n^3)$, and more precisely, $\Theta(n^3)$.*

Similarly, we can count the number of times a cell c equals some value x given n in Theorem A.2.

Theorem A.2. *$C(x; n) = \binom{n-x+2}{2}$, where $x \in \{0, 1, \dots, n\}$.*

Proof. We use the stars and bars method to count the number of ways to distribute $n - x$ samples into three cells. □

B Metric Definitions

In this section, we present a detailed overview of the classification metrics used throughout the paper. These metrics are categorized into three groups: Binomial Metrics, Joint Ratio Metrics (JRM), and Other Metrics, including fairness and specialized measures. Each table provides the formula, abbreviation, and a brief description of the metric to clarify their definitions and usage.

We repeat the definition of a binary confusion matrix for convenience in Definition B.1.

Definition B.1. *The binary confusion matrix, CM, summarizes the performance of a binary classifier and is defined as a quadruple of (TP, FP, FN, TN) .*

where:

- **TP** (True Positives): *Count of actual positives correctly predicted.*
- **FP** (False Positives): *Count of actual negatives incorrectly predicted as positive.*
- **FN** (False Negatives): *Count of actual positives incorrectly predicted as negative.*
- **TN** (True Negatives): *Count of actual negatives correctly predicted.*

Each of these counts is referred to as a cell c .

Binomial Metrics

Binomial metrics, such as Accuracy and Prevalence, are fundamental to assessing the overall performance of a classification model. These metrics are based on binomial distributions and provide general insights into model behavior across various datasets. Refer to Table 3 for a complete breakdown of these metrics.

Joint Ratio Metrics (JRM)

The Joint Ratio Metrics (JRM) are metrics derived from the confusion matrix, providing insight into the relationships between different types of classification errors. These metrics include widely used measures such as True Positive Rate (TPR), False Positive Rate (FPR), and Positive Predictive Value (PPV). A comprehensive list of JRMs, their formulas, and descriptions can be found in Table 4.

Other Metrics

In addition to the JRMs and Binomial Metrics, we also include specialized metrics such as Briscoe and Gebremedhin’s (2024) Objective Fairness Index, Matthews Correlation Coefficient (Matthews, 1975), and Prevalence Threshold. These metrics extend beyond basic performance evaluation and focus on more complex aspects, including fairness and correlation in imbalanced datasets. See Table 5 for further details.

C Proofs for Undefined Cases

In this section, we provide details and proofs for the undefined cases in Table 1 from Section 4. See Section B for metric definitions.

Theorem C.1. *Joint Ratio Metrics have $n + 1$ holes for each metric.*

Proof. With form $c_i/(c_i + c_j)$, these metrics are only undefined when $c_i + c_j = 0$ (i.e. one row or column sums to zero). For example, consider matrices where the first row sums to zero (i.e. $TP + FN = 0$). For these specific matrices, the metrics $TPR = TP/(TP + FN)$ and $FNR = FN/(FN + TP)$ are undefined. This implies that the other entries TN & FP must sum to n . So, picking TN to be any non-negative integer forces $FP = n - TN$. Since we are restricted to choosing $TN = 0, 1, 2, \dots, n$, there are $n + 1$ options for the pair TN and FP . As such, there are $n + 1$ holes for the metrics TPR and FNR .

More examples include the cases where the second row sums to zero (holes at FPR & TNR), the first column sums to zero (holes at PPV & FDR), and the second column sums to zero (holes at NPV & FOR). If we

assume c_k and c_ℓ are the entries in the zero row or column while c_i and c_j are the others, then we can count using the same restriction from above, $c_i = n - c_j$. Since there are $n + 1$ options for the pair c_i and c_j , there are $n + 1$ holes for the metric undefined when $c_k + c_\ell = 0$. Consequently, each JRM has $n + 1$ matrices at which they will be undefined. $\square$

Theorem C.2. *The original F_1 Score has $\binom{n+2}{2}$ holes.*

Proof. This score relies on the inverse of precision and recall being defined. As such, it is undefined when $TP = 0$. By Theorem A.2, some CM cell (e.g., TP) has $\binom{n+2}{2}$ counts of zero over all possible CM configurations given n . $\square$

Theorem C.3. *The simplified F_1 Score has one hole.*

Proof. Introduced as the harmonic mean of precision and recall, modern ML benchmarks often use the simplified version: $2TP/(2TP + FP + FN)$. Here, F_1 is only undefined when $2TP + FP + FN = 0$, giving 1 hole. $\square$

Theorem C.4. *The Matthews Correlation Coefficient has $4n$ holes.*

Proof. Defined as $\frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$, MCC is only undefined when any term in the denominator’s product is zero (e.g., $TP + FN = 0$). From the JRM proof, there are $n + 1$ configurations for $c_i + c_j = 0$. However, there are four overlapping cases, all when some $c_i = n$. Hence, there are $4(n + 1) - 4 = 4n$ holes. $\square$

Theorem C.5. *The Prevalence Threshold has at least $2n + 2$ holes and less than $e^\gamma n \log \log n + \frac{0.6483n}{\log \log n}$ holes for all $n \geq 3$.*

Note: Since e is Euler’s number and γ is the Euler-Mascheroni constant, $e^\gamma \approx 1.78$.

Proof. Defined as $\frac{\sqrt{TPR \cdot FPR} - FPR}{TPR - FPR}$, PT is undefined if $TP + FN = 0$ (TPR), or $FP + TN = 0$ (FPR), or $TPR - FPR = 0$. TPR and FPR are JRM metrics, hence have $n + 1$ holes each. We now move to $TPR - FPR = TP/(TP + FN) - FP/(FP + TN) = 0$.

$$\begin{aligned} & TP/(TP + FN) - FP/(FP + TN) = 0 \quad (16) \\ \iff & TP(FP + TN) - FP(TP + FN) = 0 \quad (17) \\ \iff & TP \cdot TN - FP \cdot FN = 0 \quad (18) \end{aligned}$$

Equation 18 only occurs if one row or column is a multiple of the other. For a matrix with a zero row or column, this is the set of MCC holes described above.

Table 3: Binomial Metrics: General performance metrics using a binomial distribution of outcomes, commonly used for basic model evaluation.

Metric	Abbreviation	Formula	Description
Binomial Metric	N/A	$\frac{c_i + c_j}{n}$	A ratio of cells c_i and c_j to the total count n .
Accuracy	ACC	$\frac{TP + TN}{n}$	Proportion of correct predictions to all predictions.
Prevalence	PREV	$\frac{TP + FN}{n}$	Proportion of actual positive instances in the population. (Synonym: P)
Predicted Positive Rate	PPR	$\frac{TP + FP}{n}$	Fraction of instances predicted as positive by the classifier. (Synonyms: $\hat{P}$, PP)
Inaccuracy	INACC	$\frac{FP + FN}{n}$	Fraction of incorrect predictions out of total instances. (Synonyms: Error Rate, Misclassification Rate)
Negative Prevalence	NPREV	$\frac{TN + FP}{n}$	Proportion of actual negative instances in the population. (Synonym: Complement of Prevalence, N)
Predicted Negative Rate	PNR	$\frac{TN + FN}{n}$	Proportion of predicted negative instances in the population. (Synonyms: $\hat{N}$, PN)

Table 4: Joint Ratio Metrics (JRM): Metrics that describe the relationships between confusion matrix components as ratios, commonly used to assess various types of classification performance errors.

Metric	Abbrev.	Formula	Description
Joint Ratio Metric	JRM	$\frac{c_i}{c_i + c_j}$	A ratio between a cell c_i and the sum of c_i and another cell c_j .
True Positive Rate	TPR	$\frac{TP}{TP + FN}$	Proportion of actual positives correctly identified. (Synonyms: Sensitivity, Recall, Hit Rate)
False Positive Rate	FPR	$\frac{FP}{FP + TN}$	Proportion of actual negatives incorrectly classified as positive. (Synonym: Fall-Out)
True Negative Rate	TNR	$\frac{TN}{TN + FP}$	Proportion of actual negatives correctly identified. (Synonym: Specificity)
False Negative Rate	FNR	$\frac{FN}{FN + TP}$	Proportion of actual positives incorrectly classified as negative. (Synonym: Miss Rate)
Positive Predictive Value	PPV	$\frac{TP}{TP + FP}$	Proportion of predicted positives that are actual positives. (Synonym: Precision)
Negative Predictive Value	NPV	$\frac{TN}{TN + FN}$	Proportion of predicted negatives that are actual negatives.
False Discovery Rate	FDR	$\frac{FP}{FP + TP}$	Proportion of predicted positives that are false positives.
False Omission Rate	FOR	$\frac{FN}{FN + TN}$	Proportion of predicted negatives that are false negatives. (Synonym: False Reassurance Rate)

Table 5: Other Metrics: Specialized metrics focused on fairness, correlation, and balancing different forms of classification errors.

Metric	Abbrev.	Formula	Description
Original F ₁ Score	F ₁	$2 \div \left(\frac{TP + FP}{TP} + \frac{TP + FN}{TP} \right)$	Harmonic mean of precision and recall (simplified). (Synonyms: F ₁ Measure, Dice-Sørensen Coefficient)
Simplified F ₁ Score	F ₁	$\frac{2 \cdot TP}{2 \cdot TP + FP + FN}$	Simplified version of the Original F ₁ Score.
Matthews Correlation Coefficient	MCC	$\frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$	Balanced metric accounting for all confusion matrix values, robust for imbalanced data. (Synonyms: Phi Coefficient, ϕ , r_ϕ)
Prevalence Threshold	PT	$\frac{\sqrt{TPR \cdot FPR} - FPR}{TPR - FPR}$	Threshold at which the positive prediction rate balances misclassification rates.
Marginal Benefit	$\mathcal{B}$	$\frac{FP - FN}{n}$	Considers objective testing principles in law; represents the benefit gained or lost (cost) for a group.
Objective Fairness Index	OFI	$\mathcal{B}_1 - \mathcal{B}_2$	The disparity between two subgroups' marginal benefits.
Treatment Equality	TE	$\frac{FN_1}{FP_1} - \frac{FN_2}{FP_2}$	Compares false negatives and false positives between two subgroups.

Otherwise, there is at most one multiple of that column such that the sum of the entries is n . For example, if the predicted-negative column is an integer (k) multiple of the predicted-positive column, then we have $k \cdot (TP+FN)=n$ and hence $(TP+FN)|n$. Thus, we can bound the number of holes using the sum of divisor function. Then, by Robin's Theorem (Robin, 1984), there are strictly less than $e^\gamma n \log \log n + \frac{0.6483n}{\log \log n}$ holes for $n \geq 3$.

Furthermore, since the total holes from FPR and TPR equals $2n + 2$, the lower bound is $\Omega(n)$. $\square$

Theorem C.6. *The **Treatment Equality** metric has $\binom{n_1+2}{2} + \binom{n_2+2}{2} - 1$ holes.*

Proof. Recall that TE is defined as the difference of the ratios $\frac{FN_1}{FP_1} - \frac{FN_2}{FP_2}$ for two subgroups: 1, 2. The metric is undefined when either $FP_1 = 0$ or $FP_2 = 0$. For each subgroup, the number of ways for $FP_i = 0$ is $\binom{n_i+2}{2}$, where n_i is the sample size of subgroup i . Summing over both subgroups and subtracting the case where both $FP_1 = FP_2 = 0$ (to avoid double-counting), the total number of holes is $\binom{n_1+2}{2} + \binom{n_2+2}{2} - 1$. $\square$

D Marginal Benefit's CDF

In this section, we derive the probability mass function (PMF in Theorem D.1) and CDF (Theorem D.2) of the Marginal Benefit metric, $\mathcal{B}$. Understanding this distribution allows us to compute the exact probability of observing a given value of $\mathcal{B}$ under the reference distribution.

Proof of Theorem D.1. Recall that $\mathcal{B}$ can be expressed as the average of n i.i.d. random variables X_i : $\frac{1}{n} \sum_{i=1}^n X_i$. For

$$X_i = \begin{cases} +1 & \text{with probability } p_+ = p_{FP} \\ -1 & \text{with probability } p_- = p_{FN} \\ 0 & \text{with probability } p_0 = p_{TN+TP} \end{cases} \quad (21)$$

We proceed with the proof in two parts. First, we determine the bounds, and then we derive the corresponding multinomial probability.

The Bounds of the PMF:

Let n_{+1} be the number of times $X_i = +1$ (number of false positives), n_{-1} be the number of times $X_i = -1$ (number of false negatives), and n_0 be the number of times $X_i = 0$ (number of true positives and true

Theorem D.1 (PMF of $\mathcal{B}$). *We find the PMF of $\mathcal{B} = \frac{k}{n}$ by summing over the probabilities of all counts FP and FN such that $FP - FN = k$.*

$$P\left(\mathcal{B} = \frac{k}{n}\right) = \sum_{n_{-1}=n_{-1}^{\min}}^{n_{-1}^{\max}} \left(\underbrace{\frac{n!}{(k+n_{-1})! n_{-1}! (n-k-2n_{-1})!}}_{\text{Multinomial Coefficient}} \times \underbrace{p_{\text{FP}}^{k+n_{-1}} p_{\text{FN}}^{n_{-1}} p_0^{n-k-2n_{-1}}}_{\text{Joint Probability Term}} \right) \quad (19)$$

where: $n_{-1}^{\min} = \max(0, -k)$, $n_{-1}^{\max} = \left\lfloor \frac{n-k}{2} \right\rfloor$, and k ranges over the integers $-n$ and n since $k = FP - FN \in [-n, n]$.

Theorem D.2 (CDF of $\mathcal{B}$). *The cumulative distribution function of $\mathcal{B}$ is:*

$$P(\mathcal{B} \leq b) = P(S \leq nb) = \sum_{k=-n}^{\lfloor nb \rfloor} \underbrace{\sum_{n_{-1}=n_{-1}^{\min}(k)}^{n_{-1}^{\max}(k)} \left(\frac{n!}{(k+n_{-1})! n_{-1}! (n-k-2n_{-1})!} \times p_{\text{FP}}^{k+n_{-1}} p_{\text{FN}}^{n_{-1}} p_0^{n-k-2n_{-1}} \right)}_{\text{Theorem D.1}} \quad (20)$$

where b is a real number ranging from -1 to 1 and k is an integer ranging from $-n$ to $\lfloor nb \rfloor$.

negatives). Then:

$$\begin{aligned} n &= n_{+1} + n_{-1} + n_0 \\ S = k &= n_{+1} \cdot (+1) + n_{-1} \cdot (-1) + n_0 \cdot 0 = n_{+1} - n_{-1} \end{aligned}$$

We highlight $k = n_{+1} - n_{-1}$ from above. (22)

It follows that $n_{+1} = k + n_{-1}$. We use this to redefine n_0 in terms of n , k , and n_{-1} :

$$n_0 = n - (k + n_{-1}) - n_{-1} = n - k - 2n_{-1}. \quad (23)$$

Our bounds are constrained by all cells being non-negative. As such, the counts n_{+1} , n_{-1} , and n_0 must satisfy:

$$\begin{aligned} n_{+1} \geq 0 &\implies k + n_{-1} \geq 0 \implies n_{-1} \geq -k \\ n_{-1} \geq 0 \\ n_0 \geq 0 &\implies n - k - 2n_{-1} \geq 0 \implies n_{-1} \leq \frac{n-k}{2} \end{aligned}$$

Hence, the valid range for n_{-1} is:

$$n_{-1}^{\min} = \max(0, -k), \quad n_{-1}^{\max} = \left\lfloor \frac{n-k}{2} \right\rfloor \quad (24)$$

The Multinomial Probability:

By definition, the multinomial probability of observing counts (n_{+1}, n_{-1}, n_0) is:

$$P(n_{+1}, n_{-1}, n_0) = \underbrace{\frac{n!}{n_{+1}! n_{-1}! n_0!}}_{\text{Multinomial Coefficient}} \times \underbrace{p_{\text{FP}}^{n_{+1}} p_{\text{FN}}^{n_{-1}} p_0^{n_0}}_{\text{Joint Probability Term}} \quad (25)$$

Then by substituting $n_{+1} = k + n_{-1}$ (Equation 22) and n_0 with Equation 23 into $P(S = k)$, we have:

$$P(S = k) = P\left(\mathcal{B} = \frac{k}{n}\right).$$

This concludes the proof of Theorem D.1's formula for $\mathcal{B}$'s PMF. $\square$

Proof of Theorem D.2. We start by noting that $\mathcal{B} = \frac{S}{n}$, where $S = FP - FN = \sum_{i=1}^n X_i$ is the sum of independent random variables X_i defined in Theorem D.1 (Equation 21). Furthermore, we reframe $P(\mathcal{B} < b)$ as $P(\mathcal{B} \leq b) = P\left(\frac{S}{n} \leq b\right) = P(S \leq nb)$.

Since $S = FP - FN$ is an integer ranging from $-n$ to n , the possible values of $\mathcal{B}$ are $\frac{k}{n}$ for integer $k \in [-n, n]$. Thus, $\mathcal{B}$'s CDF is:

$$P(\mathcal{B} \leq b) = \sum_{k=-n}^{\lfloor nb \rfloor} P(S = k).$$

$\square$

E Joint Ratio Metrics' CDF

In this section, we provide the full proof for Theorem 5.3 which gives the CDF of Joint Ratio Metrics (JRM). For simplicity in the following discussion, we define $k_{i+j} = k$ and $p_{i+j} = p$.

Theorem 5.3 (Restated). *The cumulative probability for a JRM score S_{obs} is given by:*

$$\begin{aligned} P(S \leq S_{\text{obs}}) &= \sum_{k=1}^n P(c_{i+j} = k) P\left(\frac{k_i}{k} \leq S_{\text{obs}} \mid c_{i+j} = k\right) \\ &= \sum_{k=1}^n \left(\binom{n}{k} p^k (1-p)^{n-k} \cdot \sum_{k_i=0}^{k_i^{\max}} \binom{k}{k_i} \theta^{k_i} (1-\theta)^{k-k_i} \right) \end{aligned} \quad (26)$$

Proof. From our analysis in binomial metrics, we have c_{i+j} and c_i following binomial distributions:

$$c_{i+j} \sim \text{Binomial}(n, p_{i+j}), \quad c_i \sim \text{Binomial}(n, p_i).$$

Since S is only defined for $k_i + k_j = k > 0$, we focus on $k \geq 1$. We begin with the simplest solution in Equation 27 which sums all probabilities for scores lower than S_{obs} .

$$P(S \leq S_{\text{obs}}) = \sum_{k=1}^n \sum_{k_i=0}^n \begin{cases} P(k_i | c_{i+j} = k) & \text{if } \frac{k_i}{k} \leq S_{\text{obs}} \\ 0 & \text{otherwise.} \end{cases} \quad (27)$$

By definition, if c_i is sampled, then c_{i+j} is also sampled, implying that $P(c_{i+j} | c_i) = 1$. Using Bayes' rule, we can express:

$$\theta = P(c_i | c_{i+j}) = \frac{p_i}{p_{i+j}}.$$

Consequently, we model the conditional sampling from a JRM M as:

$$c_i | c_{i+j} \sim \text{Binomial}(c_{i+j}, \theta).$$

We now proceed to derive the PMFs to construct the CDFs. Since the events are binomially distributed, the PMFs are given by:

$$P(c_{i+j}) = \binom{n}{c_{i+j}} p_{i+j}^{c_{i+j}} (1-p_{i+j})^{n-c_{i+j}} \quad (28)$$

$$P(c_i | c_{i+j}) = \binom{c_{i+j}}{c_i} \theta^{c_i} (1-\theta)^{c_{i+j}-c_i} \quad (29)$$

Then, the CDF of $\text{Binomial}(c_{i+j}, \theta)$ can be expressed as:

$$\begin{aligned} P\left(\frac{k_i}{k} \leq S_{\text{obs}} \mid c_{i+j} = k\right) &= P\left(k_i \leq S_{\text{obs}} \cdot k \mid c_{i+j} = k\right) \\ &= \sum_{k_i=0}^{k_i^{\max}} \binom{k}{k_i} \theta^{k_i} (1-\theta)^{k-k_i} \end{aligned} \quad (30)$$

where $k_i^{\max} = \lfloor S_{\text{obs}} \cdot c_{i+j} \rfloor$.

By expanding the PMF of c_{i+j} and combining it with the CDF of $c_i | c_{i+j}$, we prove Theorem 5.3. $\square$

F Cross-Prior Smoothing

We expand upon CPS with experiments and the proof for the bias-variance tradeoff.

F.1 Experiments

We conduct extensive experiments across fifteen commonly used classification metrics, including the eight JRMs, accuracy, prevalence, predicted positive rate, marginal benefit, MCC, F_1 score, and prevalence threshold. The experiments are performed on two popular fairness datasets: the COMPAS dataset and the Folktable Income dataset. For each metric we test 1.45 billion samples over the datasets, as described in Section 6.1.

We compare the performance of the original metrics, metrics with additive smoothing (“bashful” smoothing with $\varepsilon = 1 \times 10^{-10}$ and the non-informative prior $\varepsilon = 1$), and metrics smoothed with CPS and the prior scales $\lambda = 5$, $\lambda = 10$, and $\lambda = 20$.

F.2 Results and Analysis

Our results show that CPS consistently improves the reliability of the classification metrics, particularly for small sample sizes. Figures 5 and 6 illustrate the Mean-Squared Error (MSE) of various metrics as a function of the sample size, comparing the original metrics, metrics with additive smoothing, and CPS with different λ values.

Our experimental analyses excluded undefined values, which could make direct comparisons between CPS and the original metrics (which often include such undefined cases) potentially unfair. Nevertheless, it is important to maintain scores that closely resemble the original metrics. Therefore, we set $\varepsilon = 1e-10$ in Section 6.1. By introducing CPS, we improve over the original metrics when undefined values arise, as illustrated in Figure 5. Furthermore, Figure 5 also demonstrates that CPS performance improves when increasing λ from 5 to 20.

However, in the case of metrics that are well defined even with zero counts (e.g., binomial metrics and marginal benefit), bashful smoothing has minimal impact on error rates, as shown in Figure 6. CPS similarly improves reliability in these metrics.

We also explore the effect of the most common non-informative prior, $\varepsilon = 1$. While this approach reduces

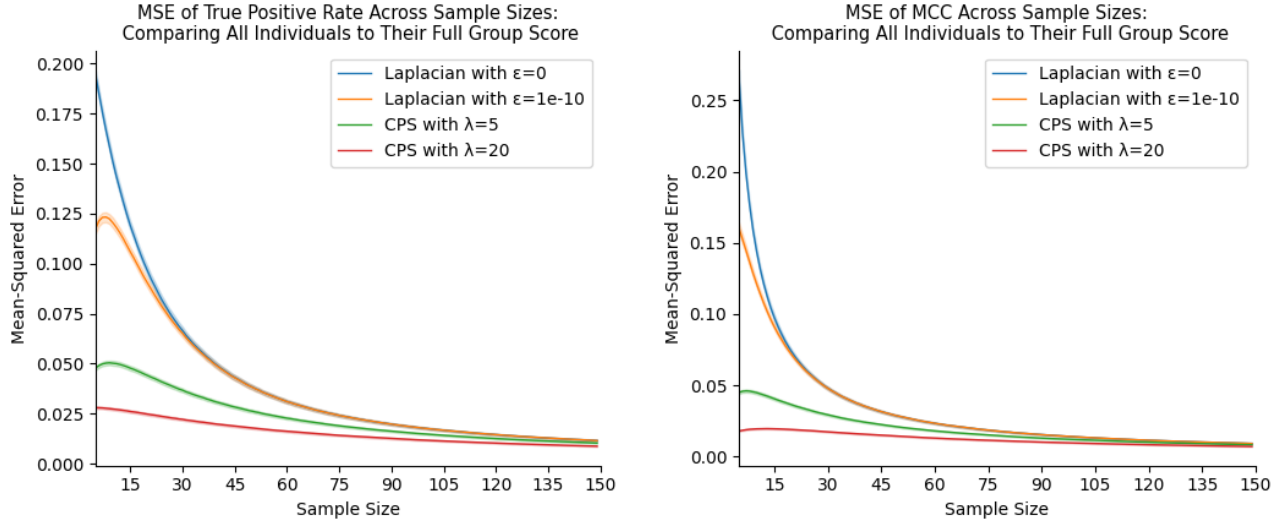


Figure 5: Effect of smoothing techniques on metrics with undefined values. Bashful smoothing ($\epsilon = 1e^{-10}$) reduces errors compared to no smoothing ($\epsilon = 0$). Cross-Prior Smoothing (CPS) offers further improvements, with the stronger prior ($\lambda = 20$) outperforming the weaker prior ($\lambda = 5$), indicating that CPS uses sufficiently informative priors.

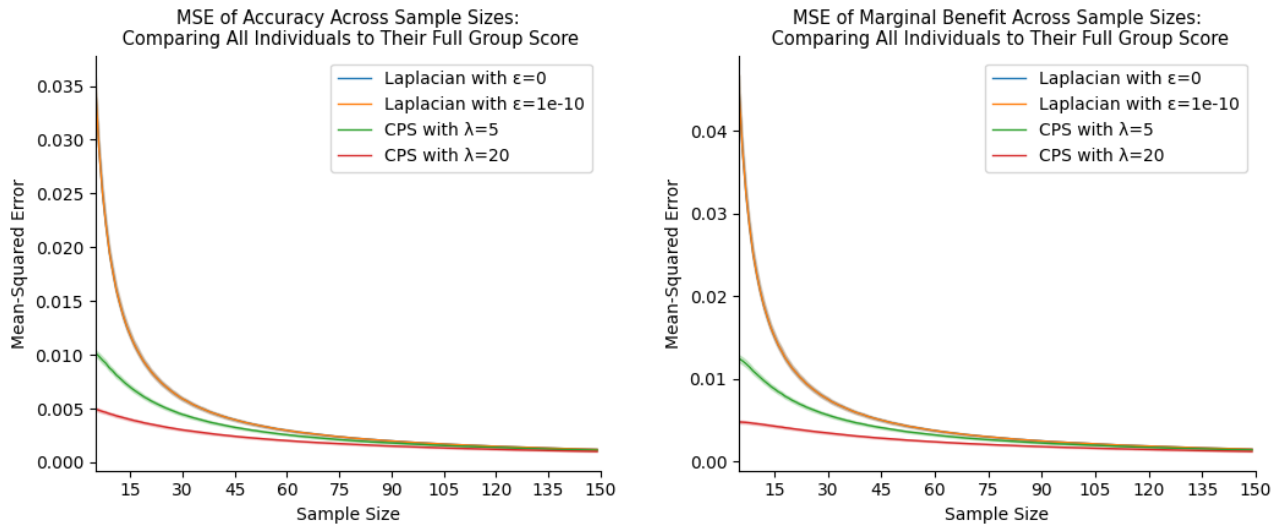


Figure 6: Comparison of smoothing effects on metrics without holes. Results show that applying a small smoothing factor ($\epsilon = 1e^{-10}$) has minimal impact compared to no smoothing. CPS continues to reduce errors, following the same trend observed in Figure 5.

errors for some metrics, it leads to inconsistent and less predictable results as shown in Figure 7. In some cases, smoothing with $\varepsilon = 1$ performs worse than no smoothing (e.g., NPV, FOR, and prevalence). In other cases, it performs better than CPS with $\lambda = 10$ for certain metrics (e.g., FNR and TPR). However, CPS with higher λ values still outperforms additive smoothing with $\varepsilon = 1$ overall. Our experiments confirm that the error bands (95% confidence intervals) in the plots are narrow and often indistinguishable, indicating high consistency across datasets and subgroups. The large number of samples (one million) ensures that the results are statistically significant and representative.

F.3 Bias and Variance

Theorem 6.1 (Restated). *The bias and variance of the CPS estimator are $\mathcal{O}\left(\frac{\lambda}{\lambda+n}\right)$ and $\mathcal{O}\left(\frac{n}{(n+\lambda)^2}\right)$, respectively.*

Proof of Theorem 6.1. Consider a cell with true probability p_c and reference probability p'_c .

- Without smoothing, $\hat{p}_c = C/n$.
- With CPS, $\hat{p}_c^{CPS} = (C + \lambda p'_c)/(n + \lambda)$.

Substituting $\mathbb{E}[C] = np_c$ yields

$$\mathbb{E}[\hat{p}_c^{CPS}] = (np_c + \lambda p'_c)/(n + \lambda),$$

so the bias is:

$$\text{Bias}(\hat{p}_c^{CPS}) = \mathbb{E}[\hat{p}_c^{CPS}] - p_c = \frac{\lambda}{n + \lambda}(p'_c - p_c).$$

As a binomial distribution, $\text{Var}(C) = np_c(1 - p_c)$. Since $\hat{p}_c^{CPS}$ is a sample proportion normalized by $n + \lambda$:

$$\text{Var}(\hat{p}_c^{CPS}) = \frac{\text{Var}(C)}{(n + \lambda)^2} = \frac{np_c(1 - p_c)}{(n + \lambda)^2}.$$

In summary,

$$\text{Bias}(\hat{p}_c^{CPS}) = \frac{\lambda}{n + \lambda}(p'_c - p_c), \quad (31)$$

$$\text{Var}(\hat{p}_c^{CPS}) = \frac{np_c(1 - p_c)}{(n + \lambda)^2}. \quad (32)$$

□

G Reproducing Results

To reproduce the results presented in this paper, please use Python 3.11. Follow the steps below:

1. Download the [repository](#).
2. Create a Python 3.11 virtual environment.
3. Install the required dependencies:

```
pip install -r requirements.txt
```

4. Run the main script:

```
python main.py
```

In case of any installation issues, a `pip_freeze.txt` file is provided with exact package versions. To install these versions, create a clean Python 3.11 virtual environment and use:

```
pip install -r pip_freeze.txt
```

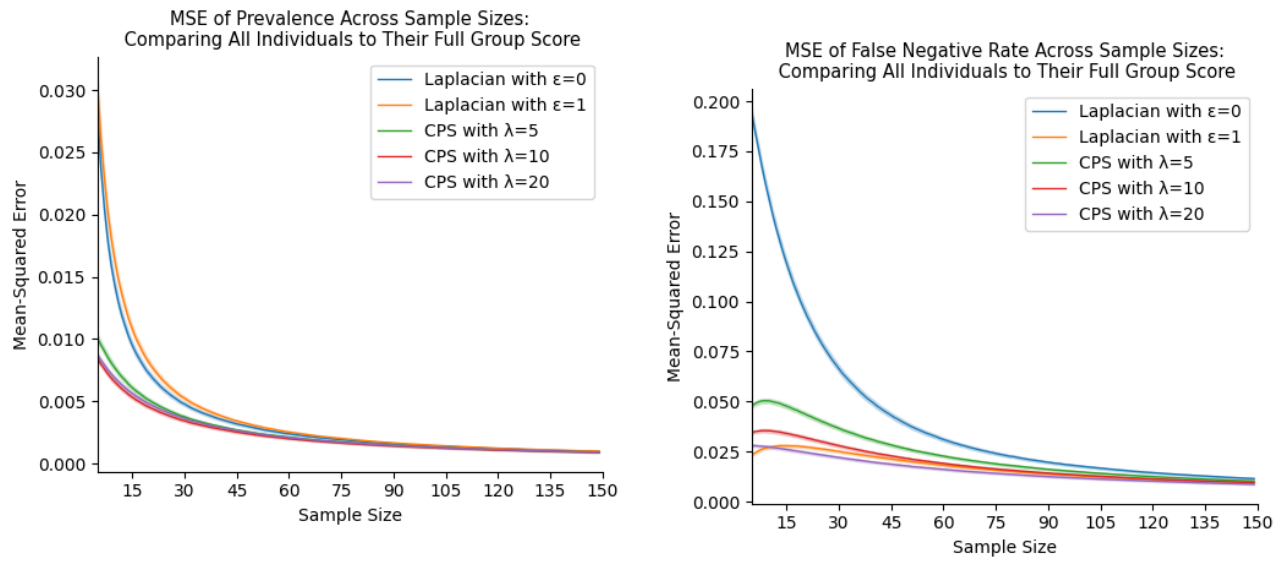


Figure 7: Smoothing with $\epsilon = 1$ yields inconsistent results. In some metrics (left), it performs worse than no smoothing, while in others (right), it outperforms CPS with $\lambda = 10$.